

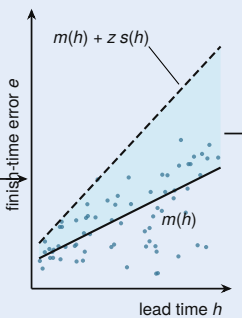
Snapshot

open tasks (r_j, d_j, w_j)

UAV states

previous plan π^-

Calibrated error model

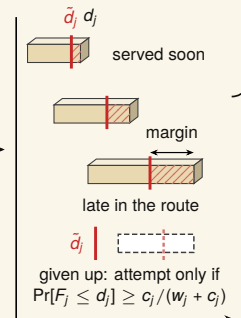


$e_j \sim \mathcal{N}\big(m(h_j),\, s(h_j)^2\big)$

$m(h) = b_0 + b_1 h,\;\; s(h) = s_0 + s_1 h$

fitted per disturbance level on development missions

Effective deadlines



$$\tilde{d}_j = t_0 + \frac{d_j - t_0 - b_0 - z s_0}{1 + b_1 + z s_1}$$

exact for a linear error model; repair and search reuse it unchanged

Anytime search

feasibility-first repair (regret-2)

ALNS destroy and repair SA acceptance

best plan π^*

no

every 5 ms: stop if $\widehat{\Delta J} \le C_R(\tau) + \mu WH$

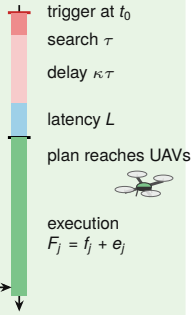
yes

$\widehat{\Delta J}$: learned gain of H more ms

C_R: expected reward lost by a later plan

hard budget $B = 250$ ms

Publish and execute



$\Pr[F_j \le d_j] \ge \Phi(z)$ for protected tasks

next call: event or $T_{\max} = 60$ s